Universität Rostock

Traditio et Innovatio

Introduction to Demographic Methods

(5: Migration)

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Introduction to Demographic Methods

1 – Introduction. Why study Demography? . . . Git, version control, course materials GitHub
2 – Mortality
3 – Fertility
4 – Population projections
5 – Migration
~~6 – Kinship~~
7 – The future, and Digital and Computational Demography
8 – Wrap up + Description of examination
9 – Final examination: 3 min presentation to share your selected topic + 2 min Q&A; 6 weeks to prepare a short essay and empirical analysis

Discussing previous week's materials

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1) Slides, Readings

1. Any questions on my slides?
2. Let's discuss the "key reading material".
3. Example: Discuss at least 1-2 points, e.g., 1) What was surprising and counterintuitive for you? 2) What did you find that was already expected? 3) What could be extended in this reading/work?
4. Any thoughts about the additional reading materials?

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2) Hands-on

1. Any problems in installing the requirements?
2. Any issues in running the codes in hands-on materials to reproduce the examples?
3. Did anyone extend the examples? How? What did you do and find?

Few points

- Migration is largely ignored in demographic methods because it can affect concepts such as generational renewal and consequently, main focus is on natural increase, at least at the start to simplify things (Wachter, 2014).
- It can change the population's age and sex structure and affect dependency ratio (To discuss Alexander's slides 2-12).
- Can it help address population decline? See UN's report "Replacement population" (Division, 2000).
- Lack of reliable data for migration has motivated many to re-purpose digital trace or "found" data to answer migration research questions (Akbaritabar, 2024; Akbaritabar et al., 2024, 2025, 2026).
- This is an area close to "Digital and Computational Demography" (Kashyap et al., 2023) which re-purposes digital traces for different demographic research questions in fertility, mortality, migration, etc.

Switch to Ernesto Amaral's slides: "Lecture11a" and "Lecture11b" and Monica Alexander's slides: "6_migration"

Hands-on part

Needed installations (or checking requirements)

For the Hands-on part of this session, I used Python. See the Readme file for this week.

Needed: Python (Vanilla or Miniforge), and virtualenv environments

The example script shows 1) how the mode-based method works, 2) how to download and use SMD data, and 3) a few example research questions that could be answered using it.

Open with Collaboratory in Google Drive:
tinyurl.com/MigMobNotebook

Where to learn basics of R and Python?

To start with Python

Check this website first:
https://www.w3schools.com/python/python_intro.asp

Check this repository by Vincent Traag and others, for an introductory course and code:
https://github.com/vtraag/intro-python

Or this one by Data Carpentry:
https://datacarpentry.org/python-ecology-lesson/

To start with R

Check this website first:
https://www.w3schools.com/r/default.asp

This "very short introduction to R" is a good start:
https://cran.r-project.org/doc/contrib/Torfs+Brauer-Short-R-Intro.pdf

Or this course by Data Carpentry:
https://datacarpentry.org/R-genomics/index.html

Reading materials for today

Key reading material, Chapter 11

- Wachter, K. W. (2014, June). Essential Demographic Methods. Harvard University Press.

Additional reading materials

- Alexander, M. (2026). Advanced Data Analysis course in Demographic Methods. Retrieved April 8, 2026, from https://github.com/MJAlexander/soc6708
- Preston, S., Heuveline, P., & Guillot, M. (2000). Demography: Measuring and Modeling Population Processes. Wiley.
- Akbaritabar, A., Dańko, M. J., Zhao, X., & Zagheni, E. (2025). Global subnational estimates of migration of scientists reveal large disparities in internal and international flows. Proceedings of the National Academy of Sciences, 122(15), e2424521122. https://doi.org/10.1073/pnas.2424521122
- Akbaritabar, A., Theile, T., & Zagheni, E. (2024). Bilateral flows and rates of international migration of scholars for 210 countries for the period 1998-2020. Scientific Data, 11(1), 816. https://doi.org/10.1038/s41597-024-03655-9

What to do before the next session?

1) Slides, Readings

1. Study my slides carefully.
2. Check the slide with reading materials.
3. Read the "key reading material".
4. Prepare at least 1-2 discussion points, e.g., What was surprising and counterintuitive for you? What did you find that was already expected? What could be extended in this reading/work?
5. Skim over the additional materials.

2) Hands-on

1. Check the slide and folder for hands-on materials.
2. Install the requirements as outlined.
3. Run the codes in hands-on materials to reproduce the examples, and if you can extend them.
4. Note down the unclear points, errors, and questions for the next session.

Thanks for your attention!

Questions and comments are always welcome!

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LinkedIn/BlueSky/Twitter/Mastodon: @akbaritabar

References

Akbaritabar, A. (2024). Thinking spatially in computational social science. EPJ Data Science, 13(1), 1–12. https://doi.org/10.1140/epjds/s13688-023-00443-0

Akbaritabar, A., Dańko, M. J., Zhao, X., & Zagheni, E. (2025). Global subnational estimates of migration of scientists reveal large disparities in internal and international flows. Proceedings of the National Academy of Sciences, 122(15), e2424521122. https://doi.org/10.1073/pnas.2424521122

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Akbaritabar, A., Theile, T., & Zagheni, E. (2026). Bilateral flows and rates of subnational and international migration of scholars worldwide by gender and field (0th ed., tech. rep. No. WP-2026-029). Max Planck Institute for Demographic Research. Rostock. https://doi.org/10.4054/MPIDR-WP-2026-029

Division, U. P. (2000). Replacement migration: Is it a solution to declining and ageing populations?

Kashyap, R., Rinderknecht, R. G., Akbaritabar, A., Alburez-Gutierrez, D., Gil-Clavel, S., Grow, A., Kim, J., Leasure, D. R., Lohmann, S., Negraia, D. V., Perrotta, D., Rampazzo, F., Tsai, C.-J., Verhagen, M. D., Zagheni, E., & Zhao, X. (2023, March). Digital and computational demography. In Research Handbook on Digital Sociology (pp. 48–86). Edward Elgar Publishing.

Wachter, K. W. (2014, June). Essential Demographic Methods. Harvard University Press.